Optimal polynomial square tests in quadratic number fields

8 November 2026

Abstract

For every quadratic extension K/F of number fields, square membership in K, with two input components in F, is expressible by one polynomial of total degree six and one auxiliary F-variable. Zero is included and no nonzero condition is needed. We prove that total degree six and degree four in the auxiliary variable are both minimal for this form of definition. The lower bound uses Hilbert irreducibility, the intermediate fields of the generic squaring map, and a valuation formula for leading coefficients. Over $\mathbb{Q}(\sqrt{2})$ we give an integer polynomial with ten monomials and coefficient height 7. A degree-ten alternative has coefficient height 8; allowing two auxiliary variables gives a quartic of height 8, or two quadratic equations. All results in this paper are unconditional. They concern square membership, rather than integrality.

1 Inputs, quantified variables, and the main results

Fix a number field F, an element $d \in F^\times \setminus F^{\times 2}$, and $K = F(\rho)$, where $\rho^2 = d$. The two input coordinates U, V describe the element $U + V\rho$ of K. Write

$$\mathcal{S}_d(F) = \{(U, V) \in F^2 : U + V\rho \text{ is a square in } K\}.$$

Throughout, zero is a square. The scalar d is fixed, not a variable.

Definition 1.1. *A polynomial square test with one auxiliary variable is a nonzero $P \in F[U, V, T]$ such that, for every $(U, V) \in F^2$,*

$$(U, V) \in \mathcal{S}_d(F) \quad \Longleftrightarrow \quad \exists T \in F \quad P(U, V, T) = 0. \tag{1}$$

Thus the polynomial has three total coordinates: two inputs and one quantified variable. All three coordinates range over the base field F. This differs from a polynomial with one input and two auxiliary variables defining $\mathbb{Z}$ over $\mathbb{Q}$.

Theorem 1.2. *For every quadratic extension of number fields as above, there is a square test of total degree six, of degree four in T, and with a nonzero constant leading coefficient in T. Every square test (1), whether or not its leading coefficient is constant, satisfies*

$$\deg P \geq 6, \qquad \deg_T P \geq 4.$$

No polynomial equation without auxiliary variables defines $\mathcal{S}_d(F)$.

The construction is proved in Sections 2 and 4; the lower bounds are proved in Section 5. Over $\mathbb{Q}(\sqrt{2})$ the following alternatives make the degree and variable tradeoff explicit:

Presentation	Auxiliary variables	Total degree	Monomials	Height
One sextic equation	1	6	10	7
One degree-ten equation	1	10	12	8
One quartic equation	2	4	8	8
Two quadratic equations	2	2	—	2

For a system, the displayed degree and coefficient height are the maxima over its equations. Coefficient height means the largest absolute integer coefficient. These are square tests; none of the displayed equations is asserted to define a ring of integers.

2 An explicit sextic over the rationals

Put

$$A = 3U^3 - 2U^2V - 3UV^2, \qquad B = -2U^3 + U^2V + 2UV^2 + V^3, \tag{2}$$

and define

$$\boxed{G(U,V,T) = 2T^4 - 2A(U,V)T^2 + B(U,V)^2.} \tag{3}$$

Lemma 2.1. *For rational U, V, one has $B(U,V) = 0$ if and only if $U = V = 0$.*

Proof. Scale a hypothetical nonzero rational zero to relatively prime integers U, V and reduce modulo 3. The values of $B(t,1)$ at $t = 0, 1, 2$ are $1, 2, 2$, respectively, and $B(1,0) = 1$ modulo 3. Thus B has no zero in $\mathbf{P}^1(\mathbf{F}_3)$, a contradiction. □

Theorem 2.2. *For every $U, V \in \mathbb{Q}$,*

$$U + V\sqrt{2} \text{ is a square in } \mathbb{Q}(\sqrt{2}) \quad \Longleftrightarrow \quad \exists T \in \mathbb{Q} \quad G(U,V,T) = 0.$$

The polynomial has total degree six, degree four in T, ten monomials, integer content one, and coefficient height 7.

Proof. Define the multiplier

$$m = (U + V) - U\sqrt{2}.$$

Direct multiplication gives

$$\mathrm{N}_{K/\mathbb{Q}}(m) = -U^2 + 2UV + V^2, \tag{4}$$

$$m^2(U + V\sqrt{2}) = A + B\sqrt{2}. \tag{5}$$

If $U + V\sqrt{2} = (u + v\sqrt{2})^2$, then

$$m(u + v\sqrt{2}) = (u^3 + 2uv^2 - 4v^3) + (-u^3 + u^2v + 2v^3)\sqrt{2}. \tag{6}$$

For a nonzero input, Lemma 2.1 gives $B \neq 0$. The product of the two components on the right of (6) is $B/2$, so neither component is zero. Taking $T = u^3 + 2uv^2 - 4v^3$ and comparing the square's components gives

$$A = T^2 + \frac{B^2}{2T^2},$$

which is exactly $G = 0$. The zero square is represented by $U = V = T = 0$.

Conversely, suppose $G(U,V,T) = 0$. If $T = 0$, Lemma 2.1 gives $U = V = 0$. Otherwise

$$\left(T + \frac{B}{2T}\sqrt{2}\right)^2 = A + B\sqrt{2}.$$

For $(U,V) \neq (0,0)$ the norm in (4) is nonzero: $(U+V)^2 = 2U^2$ would force $U = V = 0$. Division by m therefore gives the explicit square root

$$z = \frac{(U + V) + U\sqrt{2}}{-U^2 + 2UV + V^2}\left(T + \frac{B}{2T}\sqrt{2}\right). \tag{7}$$

There is no missing exceptional input: $T = 0$ was handled separately, and every nonzero input has a nonzero multiplier.

The full expansion is

$$G = 4U^6 - 4U^5V - 7U^4V^2 + 6U^2V^4 + 4UV^5 + V^6 \\ - 6U^3T^2 + 4U^2VT^2 + 6UV^2T^2 + 2T^4.$$

It proves the stated degree, monomial count, content, and height. □

The polynomial is quartic in its auxiliary variable, but it is *sextic in total degree*. For example, U^6 has degree six. No matrix entry, component, or multiplier in the displayed formula is an additional quantified coordinate; they are explicit polynomials in the two given inputs.

Proposition 2.3. *For each nonzero rational input (U, V), every algebraic root of $G(U, V, T)$ is nonzero and simple. Its discriminant in T is*

$$\operatorname{disc}_T G = 512B^2(-U^2 + 2UV + V^2)^4(U^2 - 2V^2)^2. \tag{8}$$

Proof. Taking norms in (5) gives

$$A^2 - 2B^2 = (-U^2 + 2UV + V^2)^2(U^2 - 2V^2).$$

The discriminant of $aT^4 + bT^2 + c$ is $16ac(b^2 - 4ac)^2$, obtained from its derivative and resultant. Applying this identity gives (8). Each factor is nonzero: use Lemma 2.1 and the nonsquareness of 2. The constant term B^2 is also nonzero. □

2.1 Coefficient height within the integral multiplier family

The following coefficient lower bound has a narrower scope than the degree lower bound in Theorem 1.2.

Proposition 2.4. *Let a, b, c, e be integers and put*

$$p = aU + bV, \qquad q = cU + eV, \qquad A_m = (p^2 + 2q^2)U + 4pqV, \quad B_m = (p^2 + 2q^2)V + 2pqU.$$

Divide $2T^4 - 2A_mT^2 + B_m^2$ by its positive integer content. If the resulting polynomial is a complete square test, its coefficient height is at least seven. Equality is attained by (3).

Proof. The four coefficients of B_m, in descending U-degree, are

$$2ac, \quad a^2 + 2c^2 + 2ae + 2bc, \quad 2ab + 4ce + 2be, \quad b^2 + 2e^2.$$

Soundness requires $ac \neq 0$: otherwise every $(U, 0, 0)$ would be a zero, although some rational U are not squares in K. It also requires $(b, e) \neq (0, 0)$, since otherwise every $(0, V, 0)$ would be a zero. For $V \neq 0$, the norm of $V\sqrt{2}$ is negative, so that element cannot be a square in K.

The content divides two, the coefficient of T^4. Modulo two, $B_m \equiv a^2U^2V + b^2V^3$; hence the content is two exactly when a, b are both even. In that case the primitive U^6 coefficient is $2a^2c^2 \geq 8$.

Suppose instead that the content is one and the height is at most six. The U^3T^2 coefficient $-2(a^2 + 2c^2)$ forces $a, c \in \{1, -1\}$. The V^6 coefficient $(b^2 + 2e^2)^2$ then forces

$$(b, e) \in \{(1, 0), (-1, 0), (0, 1), (0, -1)\}.$$

Write $B_j = [U^{3-j}V^j]B_m$. Then $B_0 = 2ac$ and $B_1 = 3 + 2ae + 2bc \in \{1, 5\}$. The U^5V coefficient is $2B_0B_1$, so the proposed height bound requires $B_1 = 1$. If $e = 0$, this implies $bc = -1$ and $B_2 = -2ac$; the U^4V^2 coefficient is therefore $1 - 8 = -7$. If $b = 0$, one has $ae = -1$ and $B_2 = -4ac$, giving $1 - 16 = -15$. Both conclusions contradict height at most six. Theorem 2.2 proves attainment. □

This proposition does not assert coefficient optimality among all square tests or after arbitrary rational changes of the auxiliary coordinate.

3 A degree-ten alternative

The alternative polynomial

$$P_{10}(U,V,T) = (2 - 2UT^2 + V^2T^4)(V^2 + (U - 2T^2)^2) \tag{9}$$

has total degree ten, degree eight in T, twelve monomials, and coefficient height eight. It too is a complete square test, including zero.

To prove this, suppose first that its first factor vanishes. Then $T \neq 0$ and

$$\left(\frac{1}{T} + \frac{VT}{2}\sqrt{2}\right)^2 = U + V\sqrt{2}.$$

Conversely a square root $u + v\sqrt{2}$ with $u \neq 0$ gives this factor zero with $T = 1/u$. The second factor vanishes precisely when $V = 0$ and $U = 2T^2$, because a sum of rational squares vanishes only when both squares vanish. These are exactly the inputs with a square root $T\sqrt{2}$, including zero. A product is zero if and only if one factor is zero, so the two cases use the same one quantified variable.

The highest-degree monomial is $4V^2T^8$, of total degree ten. The value eight is the largest coefficient after complete expansion; it is not an estimate for the coefficients in any integrality construction.

Over a field F of characteristic different from two, the same argument gives

$$(4 - 4UT^2 + dV^2T^4)(V^2 - c(U - dT^2)^2) = 0 \tag{10}$$

whenever d and c are nonzero nonsquares in F. One may take $c = d$. The second factor then forces $V = 0$ and $U = dT^2$.

4 A sextic over every quadratic number field

Choose $\kappa \in F^\times$ that is not a cube. Such an element exists in every number field: choose a nonzero prime ideal $\mathfrak{p}$ and $\kappa \in \mathfrak{p} \setminus \mathfrak{p}^2$. Define

$$a = U + \frac{\kappa}{2d}V, \qquad b = -\frac{\kappa}{d^2}U - \frac{1}{2}V, \tag{11}$$

$$A_d = (a^2 + db^2)U + 2dabV, \qquad B_d = (a^2 + db^2)V + 2abU. \tag{12}$$

The parameters d, κ are fixed coefficients. Thus a, b are linear and A_d, B_d are homogeneous cubics in the two input variables.

Lemma 4.1. *The cubic B_d vanishes at a pair in F^2 only at $(0,0)$.*

Proof. Expansion gives

$$4d^2B_d = -8\kappa U^3 + 6\kappa dUV^2 + (\kappa^2 + d^3)V^3.$$

If $V = 0$, its vanishing implies $U = 0$. Otherwise, put $t = 2U/V$. A zero would give an F-root of

$$X^3 - 3dX - \kappa - \frac{d^3}{\kappa}. \tag{13}$$

Let $r^3 = \kappa$. The polynomial $X^3 - \kappa$ is irreducible over F, so $[F(r) : F] = 3$. The element $s = r + d/r$ is a root of (13). Moreover, r satisfies $X^2 - sX + d = 0$ over $F(s)$. Therefore $[F(r) : F(s)]$ divides three and is at most two, so it is one. Thus s has degree three over F, and (13) is irreducible. It has no F-root. □

Theorem 4.2. *For every $U, V \in F$,*

$$(U,V) \in \mathcal{S}_d(F) \quad \Longleftrightarrow \quad \exists Y \in F \quad Y^4 - 4A_d(U,V)Y^2 + 4dB_d(U,V)^2 = 0. \tag{14}$$

This is one polynomial of exact total degree six and degree four in Y. For each nonzero input, all its algebraic roots are nonzero and simple.

Proof. Let $m = a + b\rho$. The identities

$$m^2(U + V\rho) = A_d + B_d\rho,$$
$$a^2 - db^2 = \frac{d^3 - \kappa^2}{4d^3}(4U^2 - dV^2)$$

follow by multiplication. Both factors in the norm are nonzero for nonzero (U, V): $\kappa^2 = d^3$ would make $d = (\kappa/d)^2$ a square, and $4U^2 = dV^2$ would likewise contradict the choice of d.

For a square root $z = u + v\rho$, substitution gives

$$mz = (u^3 - \kappa v^3) + \left(dv^3 - \frac{\kappa}{d^2}u^3\right)\rho.$$

The first component is nonzero when $z \neq 0$. Set Y equal to twice that component to obtain the equation in (14). Zero is represented by $U = V = Y = 0$.

Conversely, a root $Y = 0$ forces $B_d = 0$ and hence $U = V = 0$ by Lemma 4.1. For a nonzero input, $Y \neq 0$, and the displayed equation gives

$$\left(\frac{Y}{2} + \frac{B_d}{Y}\rho\right)^2 = A_d + B_d\rho.$$

Dividing the element in parentheses by m recovers a square root. At a permitted root the derivative is $8Y$ times the norm of that parenthesized element. Its squared norm is $(a^2 - db^2)^2(U^2 - dV^2) \neq 0$, so the derivative is nonzero. Finally B_d is a nonzero cubic, so $4dB_d^2$ has exact degree six; the other terms have smaller total degree and cannot cancel it. □

5 Why degree six is necessary

We give the lower bound for an arbitrary polynomial square test, without a monicity or irreducibility assumption on that polynomial. The only external theorem used in this section is Hilbert irreducibility: over a number field, finitely many irreducible polynomials over a rational function field can be specialized simultaneously so as to remain irreducible, away from a prescribed proper closed subset [1, Proposition 3.3.5 and Theorem 3.4.1].

5.1 The generic squaring map and its intermediate fields

Let u, v now be algebraically independent over F and put

$$U = u^2 + dv^2, \quad V = 2uv, \quad w = u^2 - dv^2, \quad K_0 = F(U,V), \quad L = F(u,v), \quad M = K_0(w).$$

These function fields are distinct from the fixed quadratic number field $K = F(\rho)$.

Lemma 5.1. *The extension L/K_0 has degree four. Its unique proper intermediate field is M, with $[M : K_0] = 2$. The integral closure of $F[U,V]$ in L is $F[u,v]$.*

Proof. Over $F(\rho)$ set $x = u + \rho v$ and $y = u - \rho v$. Then $U + \rho V = x^2$ and $U - \rho V = y^2$. The independent sign changes of x, y give a degree-four extension over $F(\rho)(U, V)$. The field $N = F(\rho, u, v)$ is a degree-eight normal extension of K_0: besides the sign changes, it has the automorphism interchanging x, y and sending ρ to $-\rho$. Its group is the dihedral group of order eight. The subgroup fixing L consists of the identity and this interchange. Its only proper overgroup other than itself has order four, generated by the interchange and the simultaneous sign change. Its fixed field is M, since $w = xy$ is fixed and $w^2 = U^2 - dV^2$. This proves the intermediate-field assertions.

Both u and v are integral over $F[U, V]$, as seen from

$$u^4 - Uu^2 + dV^2/4 = 0, \qquad v^4 - (U/d)v^2 + V^2/(4d) = 0.$$

Thus $F[u, v]$ is finite over $F[U, V]$. It is integrally closed, being a polynomial ring, and has fraction field L. Any element of L integral over $F[U, V]$ is also integral over $F[u, v]$, hence lies in $F[u, v]$. This proves the normalization assertion. □

Choose once and for all

$$c \in F^\times \setminus (F^{\times 2} \cup dF^{\times 2}). \tag{15}$$

Such a choice is possible by taking an element of valuation one at a prime ideal where d has valuation zero. Expanding $(a + b\rho)^2 \in F$ shows that c is a square in K if and only if c or c/d is a square in F. Thus our c is not a square in K.

Lemma 5.2. *The surface $w^2 = U^2 - dV^2$ has a Zariski-dense set of F-points whose input $U + V\rho$ is not a square in K.*

Proof. Use the rational parametrization

$$a = U + w, \quad V = b, \quad U = \frac{1}{2}\left(a + \frac{db^2}{a}\right), \quad w = \frac{1}{2}\left(a - \frac{db^2}{a}\right), \qquad a \neq 0.$$

Put $a = 2cs^2$, where $s \in F^\times$, and let $b \in F$ vary. If the resulting input were $(p + q\rho)^2$, then $p^2 - dq^2$ would equal w or $-w$. The first possibility gives $p^2 = a/2 = cs^2$, impossible. In the second, $dq^2 = a/2 = cs^2$, also impossible. The same argument includes $b = 0$. The map $(s, b) \mapsto (a, b) = (2cs^2, b)$ is dominant, and F^2 is Zariski dense. Its images are consequently dense on the surface. □

Lemma 5.3. *If P is a polynomial square test, the polynomial $P(u^2 + dv^2, 2uv, T)$ has a root $r \in L$. Every such root generates L over K_0.*

Proof. If its irreducible factors over $F(u, v)$ all had degree at least two in T, simultaneous Hilbert irreducibility would give a specialization $(u, v) \in F^2$ where none had an F-root and all required coefficient denominators were nonzero. The input is a square, contradicting the completeness of the test. A factor of degree one therefore exists.

Suppose a root r did not generate L. By Lemma 5.1, it would lie in M (which includes the possibility $r \in K_0$). Regard it as a rational function on the surface in Lemma 5.2. Away from its poles, the identity $P(U, V, r) = 0$ specializes to an F-root of P. The dense set of nonsquare inputs in that lemma meets this open subset. This contradicts the soundness of the square test. □

In particular the minimal polynomial of r over K_0 has degree four, so $\deg_T P \geq 4$. Choose its primitive polynomial representative

$$f(U, V, T) = a(U, V)T^4 + \cdots \in F[U, V, T].$$

Here "primitive" means that its coefficients as a polynomial in T have no nonconstant common factor in $F[U, V]$. Gauss's lemma implies $f \mid P$ in $F[U, V, T]$, and therefore $\deg f \leq \deg P$.

5.2 Poles and leading coefficients

For an irreducible divisor D of the affine input plane, write v_D for the integer-normalized valuation on K_0. Its extensions w_i to L are also normalized to have value group $\mathbb{Z}$; their residue degrees are denoted by f_i.

Lemma 5.4. *With the preceding notation,*

$$v_D(a) = \sum_i f_i \max(0, -w_i(r)). \tag{16}$$

Proof. For a polynomial over a valued field, the least valuation of its coefficients is additive under multiplication. To see this, divide each factor by one of its coefficients of least valuation and reduce modulo the maximal ideal; the two nonzero reduced polynomials have a nonzero product. Apply this fact over a normal closure to the monic minimal polynomial $f/a = \prod_\sigma (T - \sigma r)$. Its least coefficient valuation is $\sum_\sigma \min(0, v_D(\sigma r))$.

At a place with ramification index e_i and residue degree f_i, there are $e_i f_i$ conjugates, and their extended valuation is $w_i(r)/e_i$. The sum is therefore $\sum_i f_i \min(0, w_i(r))$. Since f is primitive, its coefficients have least v_D-valuation zero. Multiplying by a gives (16). □

Lemma 5.5. *If D is an affine F-line in the input plane, every place of L above D has residue degree at least two.*

Proof. The map associated with the finite extension $F[U, V] \subset F[u, v]$ is the squaring map. A residue-degree-one point above the generic point of D would give $u(t), v(t) \in F(t)$ such that

$$u(t)^2 + dv(t)^2 = U(t), \qquad 2u(t)v(t) = V(t),$$

where $U(t), V(t)$ are affine linear and not both constant. In $K(t)$ this would imply

$$(u(t) + \rho v(t))^2 = U(t) + \rho V(t).$$

The right side is a nonconstant linear polynomial: its leading coefficient cannot vanish because $1, \rho$ are linearly independent over F. It has a simple zero and hence cannot be a rational-function square. This contradiction proves the assertion. □

Lemma 5.6. *If $\deg f \le 5$, then a is constant and $r \in F[u, v]$ has degree at most two in u, v.*

Proof. Since $\deg_T f = 4$, its leading coefficient a has degree at most one. If it were nonconstant, its zero divisor would be an affine line D with $v_D(a) = 1$. By (16), some $w_i(r)$ would be negative. Lemma 5.5 would then make the right side of (16) at least two, a contradiction. Thus a is constant. After dividing by it, f is monic, so r is integral over $F[U, V]$ and belongs to $F[u, v]$ by Lemma 5.1.

Let $m = \deg_{u,v} r$. The four conjugates of r in the normal closure are obtained by invertible linear changes of u, v over K, so each has degree m. Their product has degree $4m$ in u, v. This product is the norm of r, the constant coefficient of its monic degree-four minimal polynomial. Because U, V are algebraically independent homogeneous quadratics, a polynomial of degree e in U, V has degree $2e$ after their substitution. Hence

$$\deg_{U,V} \mathrm{N}_{L/K_0}(r) = 2m.$$

The constant coefficient has degree at most five, giving $m \le 2$. □

5.3 The final obstruction

Theorem 5.7. *Every polynomial square test with one auxiliary variable has total degree at least six.*

Proof. Suppose $\deg P \le 5$, and choose r and f as above. By Lemma 5.6, write

$$r = q_2(u, v) + a_1 u + b_1 v + q_0,$$

where q_2 is homogeneous quadratic and $q_0 \in F$. The quadratic and constant terms are fixed by $(u, v) \mapsto (-u, -v)$, so belong to M. Since r generates L, we have $(a_1, b_1) \neq (0, 0)$.

Use the nonsquare c from (15) and specialize over an algebraic closure to

$$u = b_1\sqrt{c}, \qquad v = -a_1\sqrt{c}.$$

Then $U, V \in F$, the linear part of r vanishes, and its quadratic part has value in F. Thus the specialized value t_0 of r lies in F. The polynomial identity $f(u^2 + dv^2, 2uv, r(u, v)) = 0$ implies $f(U, V, t_0) = 0$, and divisibility $f \mid P$ gives $P(U, V, t_0) = 0$. However,

$$U + V\rho = c(b_1 - a_1\rho)^2$$

is not a square in K. The factor $b_1 - a_1\rho$ is nonzero, and c was chosen nonsquare in K. This contradicts soundness. □

Proposition 5.8. *The set $\mathcal{S}_d(F)$ cannot be defined by polynomial equations without auxiliary variables, even by a finite union of such systems.*

Proof. The squaring map is dominant: after extending scalars to K, it is the map $(x, y) \mapsto (x^2, y^2)$ in the coordinates used in Lemma 5.1. Since F^2 is Zariski dense, so is the image $\mathcal{S}_d(F)$. It is a proper subset of F^2, by the nonsquare choice (15). A finite union of sets defined by polynomial equalities is the set of F-points of a Zariski-closed set. A closed set containing a dense subset is the whole plane, which is impossible. □

These results prove all assertions of Theorem 1.2.

6 Several equations and two auxiliary variables

Corollary 6.1. *Suppose finitely many polynomial equalities, with two input variables and just one common auxiliary F-variable, define $\mathcal{S}_d(F)$. Then at least one nonzero equation has total degree at least six. The conclusion also holds for a finite disjunction of such systems.*

Proof. For a system, form the greatest common divisor of its nonzero polynomials in $K_0[T]$. If that divisor were one, a Bezout identity would exclude all common roots outside a proper closed subset of the input plane, contrary to square-input completeness. After pulling back by the squaring map and avoiding coefficient denominators, common roots are roots of this gcd. Hilbert irreducibility therefore supplies a root $r \in L$, as in Lemma 5.3. Its primitive minimal polynomial divides every nonzero equation of the system. All F-roots of that factor consequently satisfy the system, so the soundness and degree arguments of Section 5 apply unchanged.

For a finite disjunction, apply the same argument to the gcd of each branch. Branches with gcd one can occur only on a proper closed set. If every remaining pullback factor had degree at least two, one simultaneous Hilbert specialization would avoid rational roots of all branches. Thus one branch has a root in L; its minimal polynomial divides all equations in that branch, and the same contradiction applies. □

The assertion concerns polynomial *equalities*. Nonzero conditions change the situation: the quartic $4T^4 - 4UT^2 + dV^2 = 0$, together with $T \neq 0$, covers the squares with a square root whose first component is nonzero. The missing case is a second branch $V = 0$, $U = dT^2$. Replacing a nonzero condition by an equality usually requires an additional variable, which must be counted.

With two auxiliary variables there is a simpler complete presentation:

$$U = u^2 + dv^2, \qquad V = 2uv. \tag{17}$$

Over $\mathbb{Q}$, these two equations combine without further variables into

$$(U - u^2 - 2v^2)^2 + (V - 2uv)^2 = 0. \tag{18}$$

This is a quartic in four total coordinates, with coefficient height eight and eight monomials. Over a general number field F, choose a nonsquare c_0 and combine the residuals R_1, R_2 as $R_1^2 - c_0 R_2^2 = 0$; this is again a quartic with the same four coordinates. Thus allowing more variables can lower the degree, while merely adding equations does not lower the sharp bound with one auxiliary variable.

7 Relation to the elliptic and integrality constructions

For $D = (3 - \sqrt{2})^2$, direct elliptic charts use

$$x = h^2, \qquad j^2 = h^4 - D, \qquad y = hj, \qquad h = h_0 + h_1\sqrt{2}.$$

The two input components of the square-root equation are

$$U = h_0^4 + 12h_0^2h_1^2 + 4h_1^4 - 11, \qquad V = 4h_0h_1(h_0^2 + 2h_1^2) + 6.$$

They cannot both vanish. Otherwise $h^2 = \pm(3 - \sqrt{2})$; the negative sign is excluded by a real embedding and the positive sign by the nonsquare norm 7. Proposition 2.3 and the general construction in Theorem 4.2 therefore give nonzero, simple finite roots at every rational specialization of h_0, h_1. This justifies treating the corresponding root, multiplier-norm, and derivative conditions as automatic when their only purpose is to guard this chart.

The lower bound in Theorem 1.2 concerns *all* input pairs $(U, V) \in F^2$. It does not automatically give the same bound for a restricted family of radicands, or for an entire integrality construction in which different algebraic conditions interact.

In particular, the equations in this paper test square membership, not membership in $\mathbb{Z}$ or in a ring of integers. For example, $1/4$ is a square in $\mathbb{Q}(\sqrt{2})$ but is not an algebraic integer, whereas 3 is an algebraic integer and is not a square in that field. The last assertion follows by expanding $(u + v\sqrt{2})^2 = 3$: either $v = 0$ and $u^2 = 3$, or $u = 0$ and $v^2 = 3/2$, both impossible over $\mathbb{Q}$.

The ring of integers of $\mathbb{Q}(\sqrt{2})$ is $\mathbb{Z}[\sqrt{2}]$. For completeness, if $a + b\sqrt{2}$ is integral, then $2a = m \in \mathbb{Z}$ and $a^2 - 2b^2 = n \in \mathbb{Z}$. From $8b^2 = m^2 - 4n$ a reduced denominator of b has square dividing 8, so $b = k/2$ for an integer k. Then $m^2 - 2k^2 \equiv 0 \pmod 4$ forces both m and k even. Thus $a, b \in \mathbb{Z}$; the converse is immediate. Defining this ring existentially inside its fraction field is a different problem.

All auxiliary-variable lower bounds here use quantification over F. With an auxiliary variable z ranging over K, square membership already has the simple equation $z^2 = U + V\rho$ over K. Our formulas express the same condition using only one F-coordinate, instead of the two F-components of z. They do not assert a one-coordinate transfer of arbitrary existential formulas, or an integrality definition over K.

8 Exact computational data

The accompanying standard-library Python program `check_square_predicates.py` reproduces the complete coefficient tables for (3), (9), and (18). It independently checks their expanded formulas, degrees, integer content, and coefficient heights. It checks the general multiplier and norm identities by exact rational polynomial arithmetic, and computes the seven-by-seven Sylvester determinant for (8), rather than relying only on a quoted discriminant formula. A finite permutation-group enumeration verifies the overgroup calculation used in Lemma 5.1.

The generated files are

`square-predicate-polynomials.json`
`square-predicate-verification.json`.

Their calculations are finite exact certificates for the listed algebraic assertions. Hilbert irreducibility and the uniform degree lower bound are used and proved, respectively, in the text; numerical examples are not substituted for either argument.

References

[1] J.-P. Serre, *Topics in Galois Theory*, second edition, Research Notes in Mathematics, vol. 1, A K Peters, Wellesley, MA, 2007. Notes by H. Darmon. Proposition 3.3.5 and Theorem 3.4.1, pp. 24–25. https://indico.math.cnrs.fr/event/11410/attachments/4760/7351/SerreTopicsGaloisTheory.pdf.